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SILICON DETECTOR $\Delta E, E$ PARTICLE IDENTIFICATION: A THEORETICALLY BASED ANALYSIS ALGORITHM AND REMARKS ON THE FUNDAMENTAL LIMITS TO THE RESOLUTION OF PARTICLE TYPE BY $\Delta E, E$ MEASUREMENTS*

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The wide range of validity for a parameterless $\Delta E,E$ particle identification algorithm based on integrating Bragg curves using the Bethe–Bloch equation with a universal effective charge term is illustrated with detector telescope data for H through Mg. An analysis of straggling in the ΔE detector for this data is used to investigate the validity of theoretical straggling expressions which can then be used to investigate the fundamental limits to $\Delta E,E$ particle identification.

1. Introduction

The limits on resolving particle types by $\Delta E, E$ detector telescope measurements stem from three sources. First is the accuracy to which the energies involved can be measured. Second is the additional spread in ΔE values, for a given initial energy and particle type, due to energy straggling and nonuniformity of the detector. Third is the quality of the computation of Z and M from ΔE and E.

The first two limits can be subdivided into the fundamental physical limits from the detection method itself and the experimental errors due to imperfect equipment. For example, in the case of energy determination, the statistics of forming charge carriers in a silicon detector would be a fundamental physical limit while the quality of an amplifier would add a nonfundamental error. The intrinsic spread in ΔE values has a fundamental source in straggling, but it also has nonfundamental sources such as nonuniformity in detector thickness. The ultimate limits to which $\Delta E, E$ particle identification may be pushed are clearly given by the fundamental physical limits. These limits for energy resolution have been well discussed[1]) and the results may be applied directly to the question of particle resolution.

Of the two energy measurements, the ΔE resolution will have the additional component due to energy straggling of the ion passing through the detector. Although the particle identification resolution may not be limited in all regions of particle type and energy by the ΔE resolution, this additional component could cause the ΔE value to dominate over most of the regions. Therefore it is important to determine the magnitude of this contribution relative to the others and to test the theoretical predictions as a function of particle type and energy.

The straggling of charged particles in matter has been treated by several authors[2-8]). Depending upon the value of κ where

$$\kappa = \xi/\varepsilon_{\max}, \qquad \xi = \frac{2\pi e^4 z^2 NZx}{m_0 V^2},$$

and

$\varepsilon_{\max}$ = maximum possible energy transfer in a heavy particle electron collision,

one of three theories is generally appropriate. Where $\kappa \gg 1$, Bohr's[3]) expressions are valid and the energy distributions due to straggling are expected to be Gaussian. Landau[4]) studied the range where $\kappa \ll 1$. In this region the number of collisions is small and the resultant distributions are asymmetric and the most probable energy loss is followed by a long tail. The intermediate region was solved by Vavilov[5]) and no this expressions reduce to the Bohr and Landau relations at the extremes.

The value of κ can be calculated from the range of particle energies and detector thickness generally encountered in reaction studies. For the majority of cases $\kappa \ll 1$ and the distributions are predicted by Bohr to be Gaussian. However, Tschalär[6-8]), among others, studied this region in detail and found that the distributions for very large energy losses deviate somewhat from those predicted

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by the Bohr theory. The Bohr theory concerns itself with the stochastic processes but neglects the influences of the bulk shape of the energy distribution being created and propagated through the medium. When corrected for bulk effects, the distributions have a low energy tail but remain generally Gaussian. Kolata et al.[9]) showed experimentally that the corrections introduced by Tchalär were important for protons with incident energies below 20 MeV traversing a 200 μm silicon surface-barrier detector. They concluded from their data that the Tschalär corrections were required for cases where the energy losses in a ΔE detector are greater than 5%. In general they found good agreement between the distributions they measured for a number of proton energies and the distributions predicted by Tschalär.

We have extended the study of energy straggling to higher mass particles. In section 3.1 the ΔE energy resolutions for protons through ^{7}Be particles are analysed as a function of particle energy in terms of the fundamental width due to straggling and the other factors governing the resolution of Si detectors. It will be seen that in fact straggling is probably well understood by the Tschalär theory for these cases and that it is the dominant component involved in the ΔE resolution and therefore in the particle resolution.

In principle, the computational limit on particle identification can be reduced to zero but to date it has been a major problem. Short of merely defining regions in a two-dimensional matrix of ΔE versus E by the systematics of the data or of using an extensive range table look-up procedure, the computational limits from any algorithm for calculating M and Z from ΔE and E have been greater than the experimental uncertainties in the data except in very limited regions[1]). Section 3.2 describes a new algorithm based on finding and integrating the appropriate Bragg curve by using the Bethe–Bloch equation with a universal effective charge correction. While it still fails in places to reach an accuracy equal to that inherent in the data, it approaches or exceeds the experimental accuracy over a much larger region than any previous algorithm except for those based on range table look-up procedures. Remarkably, it does this with no adjustable parameters.

2. Experimental procedures[10])

The particles used in this study were generated by 200–500 MeV protons incident on a silver target located in a scattering chamber at the TRIUMF accelerator. In the resultant reactions a distribution of particle types and energies is produced[11]). These particles were detected by standard transmission Si surface barrier detectors arranged in telescopes of 2–4 data detectors followed by a detector to veto events where the range of the particle exceeded the data telescope thickness. Depending upon the telescope and particle studied, any of the leading detectors could be used as the ΔE detector with those following used as the E. Therefore every valid event represented a case where a particle had traversed ΔE detectors and stopped in E detectors with its total energy given by the sum of the ΔE and E values.

Standard linear electronic systems were used to generate energy signals which were recorded on magnetic tape event by event to 11-bit accuracy. Majority 2 logic was used to identify the true

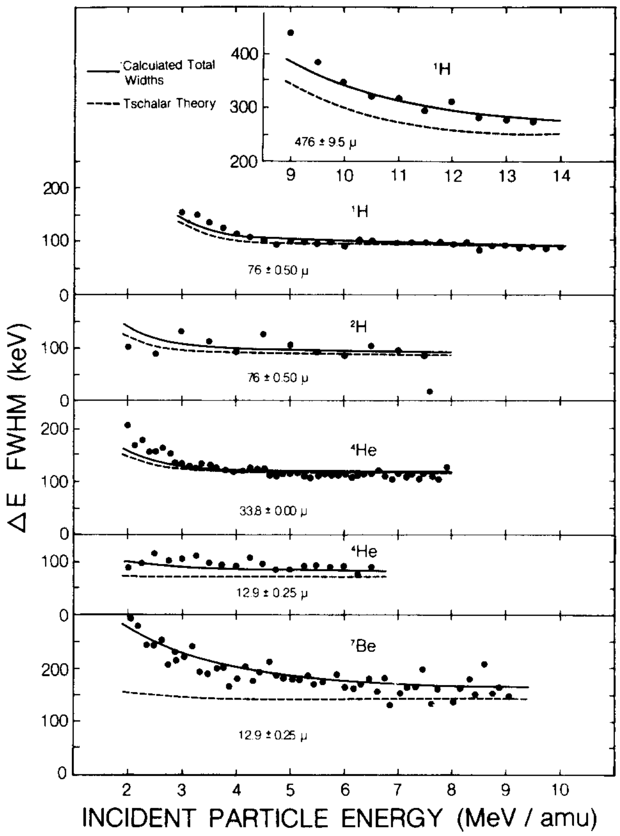

Fig. 1. Comparison of calculated ΔE energy widths with experimental values obtained with Si ΔE detectors. Dashed lines give the energy straggling contribution calculated from Tschalär expressions. Solid lines include additional contributions from window widths, detector and electronic resolution, and detector nonuniformity.

events. Each event on tape was then a series of digitized energy signals identified as to which detector generated the signal and which group of signals were due to a particular event.

Each detector was calibrated by normalizing known energy signals, generated by alpha particles from an ^{241}Am or ^{212}Pb source stopping in that detector, to a precision pulse generator. An energy calibration of the amplifier range was then determined from the pulser. The calibration points were fitted by a linear least-squares program and were generally found to be within one part in 2000 of the calculated curve from 2–98% of the energy range.

The recorded data was processed on an IBM 370 computer at the Simon Fraser University Computer Centre. An initial program checked and sorted the data of interest writing a new data tape which became the data set used by several different programs depending upon the analysis to be performed. During this initial step the data was digitally stabilized relative to pulser events collected concurrently with the particle events.

For the energy straggling study, the fwhm widths of the peaks in the ΔE detector energy distributions were determined for a thin window in incident particle energy. A program identified a peak and fitted it to Gaussian distribution. Chi-squared values were determined for several fits from 10% on either side of the initial calculated sigma and the one with the lowest chi-squared value was chosen as representing the peak and the fwhm value was calculated. Those peaks so identified and characterized which were due to single particle types were selected for the study of energy straggling.

The development of the particle identifying algorithm involved selecting the appropriate ΔE and E detectors and using the corresponding energy values in the PI calculation. A large range of particles and configurations could be studied simultaneously because of the available choice of ΔE and E detectors.

3. Results and discussion

3.1. Straggling measurements

The results of the straggling measurements are displayed in fig. 1 where the data covers a range of particles and detector thicknesses. The data for two cases, ^{1}H in a 476 μm detector and ^{4}He in a 12.9 μm detector, are the result of multiple determinations averaged together. These combined distributions had correlation coefficients of 0.85 in the ^{1}H case and 0.94 in the ^{4}He case. All of the other cases are for single determinations.

The dashed lines are the results of a calculation of the energy straggling using Tschalär's expressions[6]). To these theoretical widths must be added all other contributions which would be included in the measurements. The width will be broadened as a result of the change in dE/dx values across the finite energy windows used to determine the distributions. This effect was calculated and folded into the Tschalär values. In most cases the correction was $<1\%$ except at the very lowest energies where the effect added $<4\%$ for the ^{1}H, ^{2}H, and ^{4}He cases and $<10\%$ for the ^{7}Be case. The contributions due to electronic noise and detector charge collection characteristics were taken from the resolution measured for alpha particles in the detectors and the estimated values for protons. These corrections were added in quadrature to the corrected Tschalär values and were generally less than a 1% effect. The remainder of the ΔE widths was attributed to nonuniformity in the detector. The solid curves in fig. 1 give the results of combining all of these contributions to the ΔE resolution. The required variations in detector thicknesses are given as errors on the detector thickness and are within the specifications of the manufacturer and agree for different particles in the same detector.

All other possible contributions to the energy widths have been neglected. Multiple scattering would increase the effective detector thickness and channeling would give a low energy tail to the distributions as would the loss of charges from the detector surfaces. Fluctuations in charge pick-up by the ion would also affect the widths below 3 MeV/amu for the ions studied here. However, the data presented here does not require the addition of these other effects to give a consistent and adequate explanation of the measured widths. Within the accuracy of the measurements, the experimental widths can be explained by the sum of Tschalär's predictions of energy straggling, electronic and detector noise, and detector nonuniformity. Although the small increase in widths at the lower energies for a few of the cases might suggest the need for multiple scattering and/or charge pickup corrections, additional data at lower energies with greater statistical significance is needed to justify their inclusion.

Perhaps the most significant result of these measurements is the observation that the dominant contribution to the resolution of a ΔE detector is that due to energy straggling. For heavier particles ($A>4$) at lower energies the detector uniformity is also important, but in all cases the inherent resolution of the detector system is insignificant. Therefore, in as much as the ΔE resolution will determine the particle resolution of a ΔE, E telescope, the figure of merit for a system will be determined primarily by the fundamental phenomenon of energy straggling and secondarily by the physical uniformity of the detector. Other factors are relatively unimportant.

However, the particle resolution can be improved if two or more ΔE detectors are used and the resulting calculated PI values are required to agree to within narrow limits for the event to be kept, in effect selecting only the central parts of the energy distributions. A significant fraction of the events will be lost in such an analysis and a far higher energy cut-off of the data will be incurred. Under certain circumstances the improved particle resolution might warrant this loss of data.

3.2. Particle identification algorithm

The quality of a particle identification algorithm for ΔE, E detector telescope data depends on how well Z and M for a given type of event can be extracted from the average values of the energies deposited in the detectors for such an event. The energies, M, and Z can be related using ranges and stopping powers by the expression

$$\Delta E = \int_{R(E_0,Z,M)-t}^{R(E_0,Z,M)} S(x,Z,M)\,\mathrm{d}x\,, \tag{1}$$

where

ΔE = the energy deposited in the ΔE detector,

t = the thickness of the ΔE detector,

E_0 = the initial energy of the particle $=\Delta E+E$,

$R(E_0, Z, M)$ = the initial range of the particle in the ΔE material, and

$S(x, Z, M)$ = $-\mathrm{d}E/\mathrm{d}x$ = the rate of energy loss in the ΔE material for a particle of mass M and charge Z at a residual range x.

In short, the quality of the algorithm depends on how well one can find and integrate the correct Bragg curve.

Almost all current calculational particle identification algorithms (as opposed to range–energy table lookup procedures such as those of refs. 12 and 13) use assumptions which become invalid as the ΔE thickness approaches the range of the initial particle. However, data where the ΔE thickness is nearly equal to the particle range is of major importance in our experimental program. Hence it was decided to try obtaining M and Z from eq. (1) by using a new approach with less restrictive assumptions.

The first assumption in this new approach is based on the relatively slow rate of change in the slope of Bragg curves for residual ranges above the Bragg maximum. A two-point approximation is used in eq. (1) with the integral being approximated as the ΔE thickness times the average of the initial and final values of the stopping power. Problems with this approximation at residual ranges below the Bragg maximum can be overcome to a large extent by appropriately modifying $S(x, Z, M)$. Higher order approximations to the integral are of course possible but not straightforward since a transformation is now made from $S(x, Z, M)$ to $S(E, Z, M)$ yielding the approximate relation

$$\Delta E = \tfrac{1}{2}t\,[S(E_0, Z, M) + S(E_0-\Delta E, Z, M)]\,, \tag{2}$$

from which it is desired to extract Z and M knowing t, E_0, and ΔE.

Clearly an accurate description of $S(E, Z, M)$ is desired except at very low values of E where compensation is required for the error due to the approximate form of the integration. Since the particles of interest are non-relativistic, an approximate form of the Bethe–Bloch equation

$$-\mathrm{d}E/\mathrm{d}x = \frac{aMq^2}{E}\ln\left(\frac{bE}{M}\right) \tag{3}$$

can be used where a and b are constants which depend on the stopping medium and q is the average effective charge of the particle. At very low values of E/M, the Bethe–Bloch equation is not valid, but there are already troubles in this region. The problems at low E/M can be largely circumvented by taking

$$S(E,Z,M) = \begin{cases} -\mathrm{d}E/\mathrm{d}x(E,Z,M)\,, & \text{for } E(\mathrm{MeV}) > Z,\\ -\mathrm{d}E/\mathrm{d}x(Z,Z,M), & \text{for } E(\mathrm{MeV}) < Z, \end{cases} \tag{4}$$

which prevents $S(E, Z, M)$ from yielding an unrealistically low value for the integral in eq. (1).

Use of the rms effective charge q instead of the atomic number Z is vital. The function

$\gamma(E,Z,M) = q/Z$

has been studied by several authors[14]). Booth and Grant[15]) found it possible to prepare a universal charge curve in terms of

$$\xi = \frac{E}{MZ^{\frac{4}{3}}} \tag{5}$$

to fit their data and that of others for $0<\xi<0.125$ where ξ is calculated using E in MeV and M in amu. For our work, it is convenient to fit their data[16]) with

$$G(\xi) = \gamma^2(\xi) = 1 - \exp(-0.2-11.5\xi), \tag{6}$$

which provides a good fit for $0.02<\xi<0.125$ and extrapolates nicely to infinity. The departure of this fit from the data below $\xi=0.02$ is not important for the purposes here.

The effects of Walske's[17]) form of the correction term commonly included in the Bethe–Bloch equation to account for the nonparticipation of inner shell electrons in the energy loss process was also investigated. The effects of this correction did not noticeably improve the particle resolution and since the other approximations already employed are not sufficiently accurate to justify the increased computing time required by use of this correction term, it was not included in our final expressions.

Using eqs. (3), (4) and (6) in eq. (2) and rearranging terms yields

$$MZ^2 = \frac{2\Delta E}{at}\left\{E_0^{-1}\, G\left(\frac{E_0}{MZ^{\frac{4}{3}}}\right)\ln\left[\frac{bE_0}{M}\right] + \right.$$
$$\left. + (E_0-\Delta E)^{-1}\, G\left(\frac{E_0-\Delta E}{MZ^{\frac{4}{3}}}\right)\ln\left[\frac{b(E_0-\Delta E)}{M}\right]\right\}^{-1}, \tag{7}$$

where the energies E_0 or $E_0-\Delta E$ are to be replaced by Z when their value in MeV is less than Z. For silicon, $a=0.0166\ \mathrm{MeV^2/\mu m\cdot amu}$ and $b=12.4\ \mathrm{amu/MeV}$. Eq. (7) has the classic particle identification combination of M and Z on the left hand side but, as usual, there is still some M and Z dependence on the right hand side. With a and b chosen so that M is in amu, it can be seen that the right hand side is not very sensitive to M and will suffer little under the assumption that $M=2Z$. It is now possible to solve eq. (7) by iteration using this same assumption on the left hand side in the form

$$Z = (\tfrac{1}{2}MZ^2)^{\frac{1}{3}}.$$

One assumes an initial $Z=Z_0$, evaluates the right hand side, takes the appropriate cube root, and compares this new Z to Z_0. If they are not equal within desired limits, the new Z is used as Z_0 and the iteration continued until there is self-consistency. The final calculated particle identification value, PI, is taken to be this self-consistent value and should be approximately $(\frac{1}{2}MZ^2)^{\frac{1}{3}}$.

A program written in BASIC to calculate PI using the above assumptions is given in the appendix to this article. Self-consistency within 0.01 in PI requires twelve or fewer iterations in all cases we have studied. For the majority of our data, 4 or 5 iterations is normal. It is important to note that the output from this algorithm is a readily interpreted value approximately equal to Z for any thickness ΔE detector and that the only inputs are the ΔE detector thickness and the two energies. There are no adjustable parameters.

The performance of this particle identification algorithm can be demonstrated using data collected with the system described in section 2. This data conveniently provides a range of ΔE thickness values from 12.9 to 476 μm. Fig. 2 shows a contour plot generated by analyzing data from an experimental run using a 12.9 μm ΔE detector. A region of the resulting data set of PI and total energy values was condensed into a 100 by 100 matrix and then processed by an interpolating contour program to yield contour levels at 2^n counts/bin for $n=1$–10. The ΔE gain used resulted in signals outside of the analyzer range for fragments beyond Na. The pulser used during the run is seen in the upper right corner. This and similar plots from other runs can be used to give a reasonable idea of the quality of the algorithm (note the isolated isotope ^{7}Be). However, structure in the data from the nature of the fragmentation process and from the finite experimental widths of the distributions as well as the problem of producing readable contour plots makes it difficult to display succinctly the information concerning the PI algorithm itself. To circumvent the display problems and to provide a convenient method of combining information from many separate experimental runs, it is useful to use ridge plots where only the centroids of various particle groups are displayed as is done in figs. 3 and 4 where results for ΔE detectors of five thicknesses are shown. Lines labelled simply as an element are from the centroid of the isotopes of that element as produced by the fragmentation process being studied. Where there are sufficient statistics to define the centroid well, solid lines are used. Dashed lines imply there are sufficient data to define the exis-

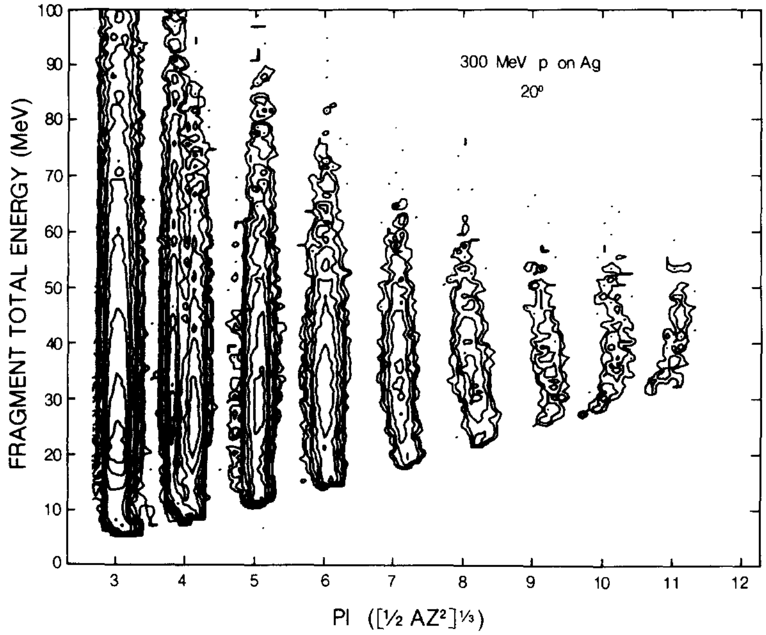

Fig. 2. Contour plot of fragment total energy vs calculated PI for data taken with a 12.9 μm Si ΔE detector. Contour levels are at 2, 4, 8, 16, ...

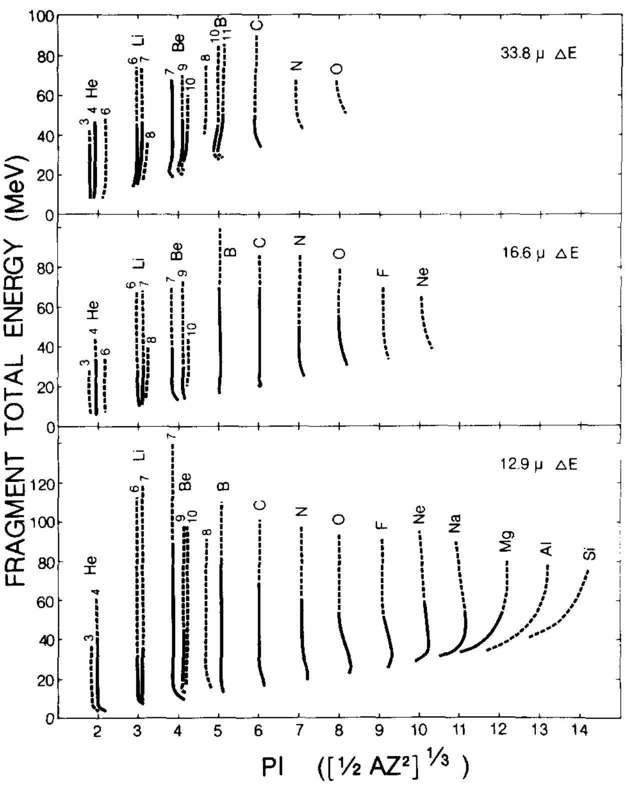

Fig. 3. Ridge plot of fragment total energy vs calculated PI for data taken with various Si ΔE detectors illustrating the quality of the PI calculation.

tence of the centroid but its position is not well defined either from a lack of sufficient counts or from interference due to a neighboring isotope. Abruptly terminated solid lines mean a range limit in the detector telescopes used has been reached – particles are either not reaching the E detectors with enough energy to give a signal in the electronics or they have passed completely through the E detectors.

In preparing these figures, it was noticed that the centroids, but not the general shapes, were sometimes shifted slightly from one group of runs to another when there had been a change in the experimental configuration. This was first noticed when the preamplifier load resistors were lowered in value so that accurate corrections could be made to maintain the rated bias voltages on the thin detectors. This was required because the detector leakage currents had increased due to radiation damage. Although this problem was not studied in detail, the fact that the cases where the bias voltage was low gave low PI values and that the correction was only required if the detector was used as a ΔE detector suggested a correction involving the energy calibration of the ΔE detectors to compensate for incomplete charge collection at the exit low field side of the detector. By renormalizing the ΔE calibration (or by changing

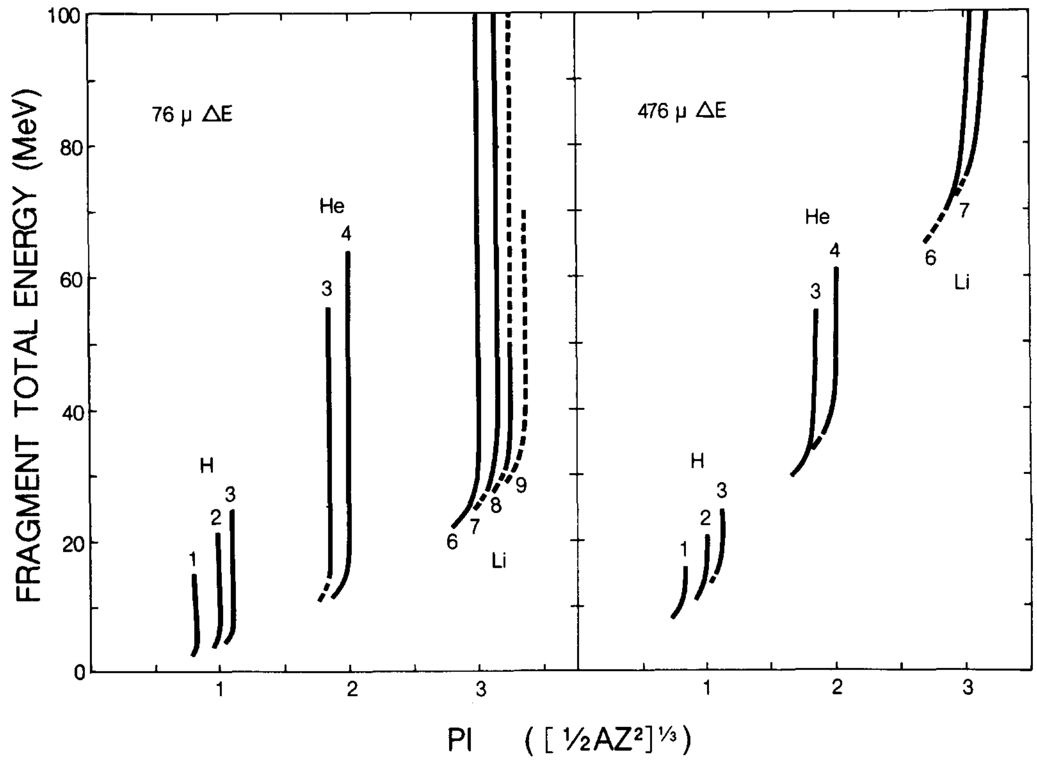

Fig. 4. Ridge plot of fragment total energy vs calculated PI for data taken with various Si ΔE detectors illustrating the quality of the PI calculation.

the detector thickness which is almost an equivalent process as far as the PI algorithm is concerned for the corrections used), good overlap of all runs was achieved. The largest renormalization used was a 12% correction which resulted in an approximately 4% correction to the PI value. The majority of the corrections were smaller or not required.

Similar plots for three other algorithms can be found in ref. 18. None of these other calculations approach the results obtained here, nor are they applicable to as wide a range of ΔE thicknesses without parameter adjustment.

The deviation toward a lower value of PI as the fragment energy becomes lower, noticeable in the thicker ΔE detectors, is believed to be due to the increasing error in the two-point approximation to the integral in eq. (1). The complex behavior at the very lowest energies probably results from a complicated mixture of the invalidities in the assumptions – especially the treatment of $S(E, Z, M)$ at low E. Deviations in the higher Z fragments undoubtedly reflect the difficulties in constructing a simple universal effective charge correction.

Fig. 5 shows a histogram of calculated PI values summed over all measured energies for an experimental run using a 12.9 μm detector with the gains in the electronics set to detect fragments with a wide range of Z values. All of the high intensity H and He fragments were rejected in the electronics as were the low energy Li, Be, and B fragments in order to conserve magnetic tape manipulations and computing time required for the data analysis. This histogram compares favorably with similar ones in refs. 11 and 19 where slightly thicker 16 and 20 μm detectors have been used. It is apparent that fig. 5 indicates much better resolution for the resolved isotopes. However, it is not possible to separate the experimental and calculational contributions to such figures from the figures themselves. It may also be misleading to compare results of elements with unresolved isotopes because the fragment production sources are different.

Since the spacing in ΔE values for adjacent isotopes with the same incident energy increases faster than the width in ΔE due to straggling, the PI resolution increases if thicker ΔE detectors are

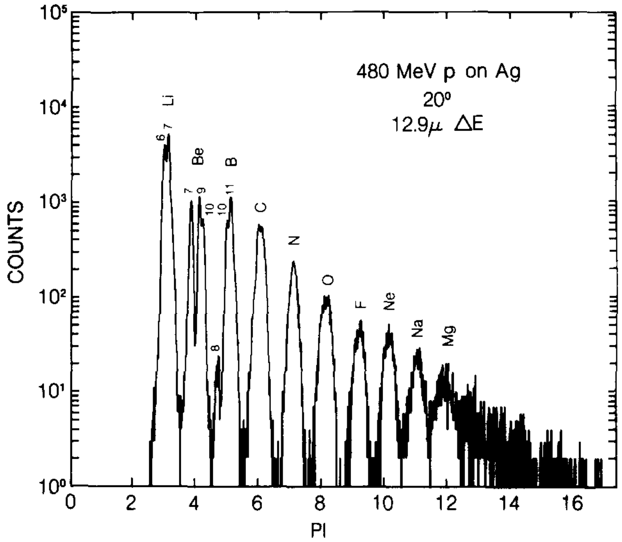

Fig. 5. PI spectrum of fragments for data taken with a 12.9 μm Si ΔE detector. H and He isotopes were electronically rejected.

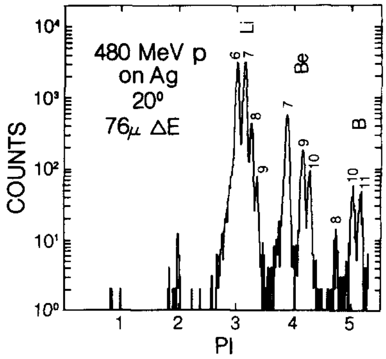

Fig. 6. PI spectrum of fragments for data taken with a 76 μm Si ΔE detector. H and He isotopes were electronically rejected.

used. If one neglects the lowest energy fragments, improved isotopic separation can be achieved with thicker ΔE detectors using the algorithm presented here. Fig. 6 shows results for a 76 μm detector where only events for which $E/(E+\Delta E)>\frac{2}{3}$ are allowed and again H and He fragments have been suppressed in the electronics. This energy requirement is not as severe as it may at first appear. For example, only the lowest 10 MeV indicated for ^{6}Li in fig. 4 is discarded by this requirement.

4. Conclusions

The results of this study have shown that the largest component to the ΔE resolution comes from the fundamental phenomenon of energy straggling provided the detectors are of reasonable uniformity. As a consequence, energy straggling will generally dominate the particle resolution that can be obtained by a ΔE–E measurement and further attempts to improve the inherent energy resolution of ΔE Si surface barrier detectors is of diminishing value. It has also been shown that it is fruitful to develop a theoretically based particle identification algorithm. With the inclusion of a universal effective charge term, an algorithm without adjustable parameters has been generated which can cover a wide range of particle energies and types.

We acknowledge with thanks the help received from the operations staff of TRIUMF in obtaining the proton beams used in this experiment. We are especially indebted to F. M. Kiely, R. Toren and A. Kurn for their help with the development of the computer programs which we have used.

Appendix

A PI calculation.

The following is a listing of a BASIC program, using the technique discussed in the text, to calculate the PI value of a particle. The only inputs are the ΔE detector thickness T1, the ΔE value E1, and the E value E2. This calculation requires agreement within 0.01 (see statement number 310) in successive iterations before printing out the PI value.

```
*LIST
0010  INPUT "DE THICKNESS (MICRONS) = ",T1
0050  INPUT "                DE (MEV) = ",E1
0060  INPUT "                 E (MEV) = ",E2
0100  REM "CALCULATE FIXED VALUES FOR THIS EVENT"
0110  LET T=60.2*E1/T1
0115  LET E0=E1+E2
0200  REM "MAKE AN INITIAL GUESS FOR Z"
0210  LET Z=2
0220  LET Z9=0
```

```
0300 REM "CALCULATE TRUE Z BY ITERATION"
0310 IF ABS(Z-Z9)<.01 THEN GOTO 9990
0320 LET Z9=Z
0340 LET Q0=E0
0350 IF Q0<Z THEN LET Q0=Z
0360 LET Q2=E2
0370 IF Q2<Z THEN LET Q2=Z
0500 REM "CALCULATE AVERAGE EFFECTIVE CHARGE TERMS"
0510 LET P0=2*(Z↑2.3333)
0520 LET P=Q0/P0
0530 LET G0=1-EXP(-.2-11.5*P)
0560 LET P=Q2/P0
0570 LET G2=1-EXP(-.2-11.5*P)
0600 REM "CALCULATE ET PART OF DENOMINATOR"
0610 LET D0=(G0/Q0)*LOG(6.2*Q0/Z)
0700 REM "CALCULATE E2 PART OF DENOMINATOR"
0710 LET D2=(G2/Q2)*LOG(6.2*Q2/Z)
1000 REM "CALCULATE NEW Z FROM ABOVE"
1010 LET Z=(T/(D0+D2))↑.3333
1090 GOTO 0300
9990 PRINT "                              PI = ",Z
*RUN
DE THICKNESS (MICRONS) = 12.9
                DE (MEV) = 20.3
                E (MEV)  = 30.4
                PI       = 8.95475
END AT 9990
```

References

1) F. S. Goulding and B. G. Harvey, Ann. Rev. Nucl. Sci. **25** (1975) 167 and references therein.
2) H. D. Maccabee, M. R. Raju and C. A. Tobias, Phys. Rev. **165** (1968) 469.
3) N. Bohr, Kgl. Danske Vid. Selsk. Mat. Fys. Medd. **18**, no. 8 (1948).
4) L. Landau, J. Phys. USSR **8** (1944) 201.
5) P. Vavilov, Sov. Phys. JETP **5** (1957) 749.
6) C. Tschalär, Nucl. Instr. and Meth. **61** (1968) 141.
7) C. Tschalär, Nucl. Instr. and Meth. **64** (1968) 237.
8) C. Tschalär and H. D. Maccabee, Phys. Rev. **B1** (1970) 2863.
9) J. J. Kolata, T. M. Amos and H. Bichsel, Phys. Rev. **176** (1968) 484.
10) A detail description of the experimental procedures will be published shortly.
11) E. K. Hyde, G. W. Butler and A. M. Poskanzer, Phys. Rev. **C4** (1971) 1759; R. G. Korteling, C. R. Toren and E. K. Hyde, Phys. Rev. **C7** (1973) 1611.
12) B. Hird, R. W. Ollerhead, Nucl. Instr. and Meth. **71** (1969) 231.
13) H. Bichsel, Nucl. Instr. and Meth. **78** (1970) 277.
14) H. D. Betz, Rev. Mod. Phys. **44** (1972) 466, and references therein.
15) W. Booth and I. S. Grant, Nucl. Phys. **63** (1965) 481.
16) The fit given in eq. (4) of ref. 15 must contain a misprint.
17) M. C. Walske, Phys. Rev. **88** (1952) 1283.
18) E. T. Chulick, J. B. Natowitz and C. Schnatterly, Nucl. Instr. and Meth. **109** (1973) 171.
19) G. W. Butler, A. M. Poskanzer and D. A. Landis, Nucl. Instr. and Meth. **89** (1970) 189.